



# Multilinguality at Common Crawl

Improving Language Coverage for the Largest Open Web Corpus

Pedro Ortiz Suarez and Laurie Burchell

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## Common Crawl – A Brief Introduction

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What Is Representative?

Size and Freshness

Crawler Politeness

Data Pipelines

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

WMDQS

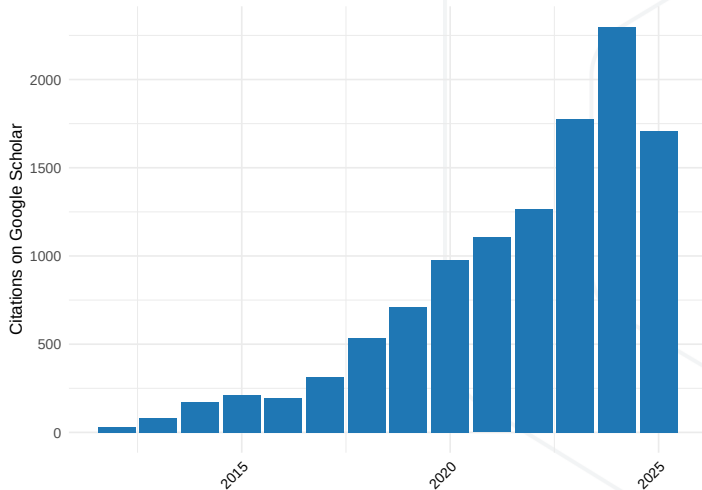
CommonLID Dataset

Summary

# About Common Crawl

- We're a non-profit making web data accessible to programmers and data scientists
- Free archive of the public web since 2007, founded by Gil Elbaz
- Hosted as Open Dataset on Amazon Web Services [1, 2]
- Cited in over 10,000 research papers

# Papers Citing Common Crawl (through 9/25)





# Common Crawl Use Cases

- Search indexes
- Sociopolitical research
- Linguistic analyses
- Internet security research
- Economic research
- AI/ML training
- A lot more...

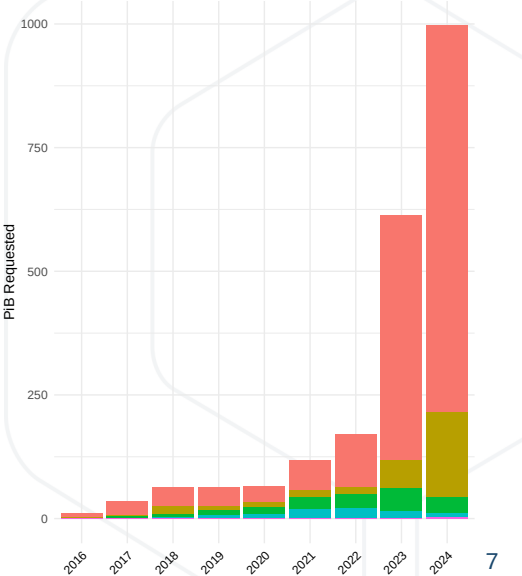
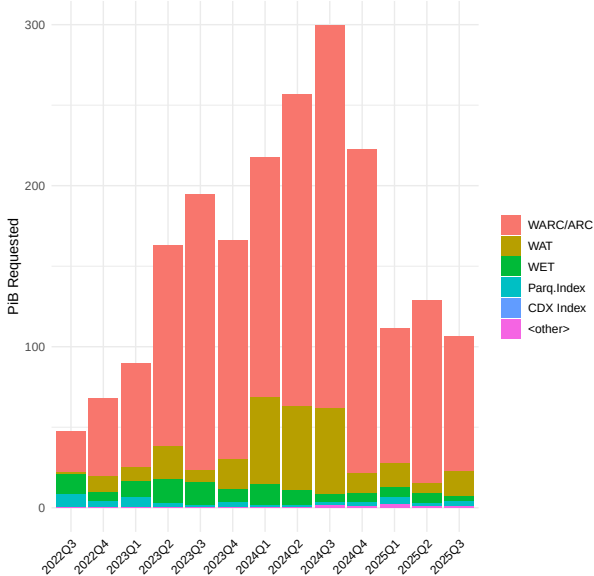
# Our Mission

- Make high-quality public web crawl data (once limited to large search engines) available to everyone
- Support people, governments, and organizations make data-driven decisions and tackle global issues
- Build an open knowledge ecosystem that encourages sharing and innovation
- Support understanding and preservation of language, community, and culture

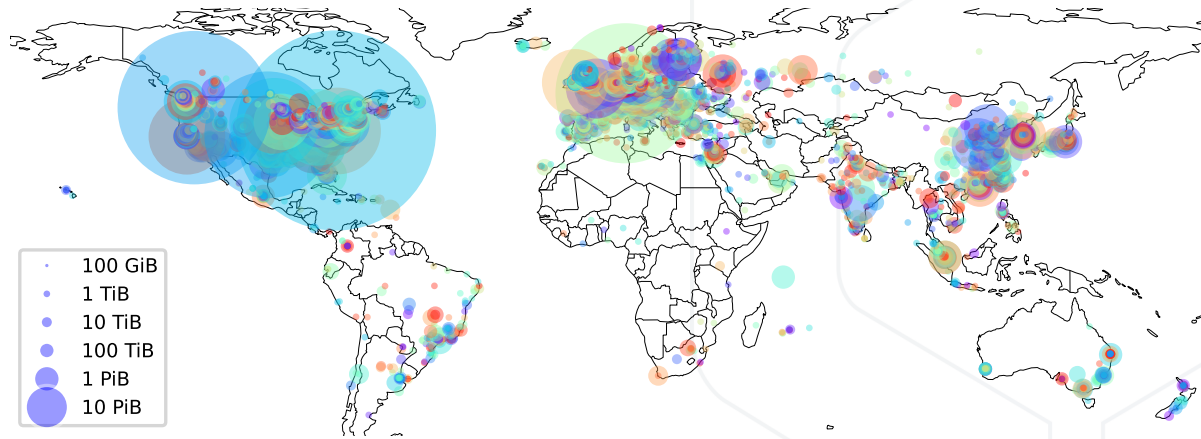
# Data Overview

- Over 300 billion web pages spanning 17 years (2008 – 2025)
- Around 2.5 billion pages added each month
- More than 100 crawl archives released to date
- 10.1 PiB of data (Sept 2025)
- Additional data products: web graph, host index, etc.

# Data Downloads



# Downloads by Geographical Location



# Data Collection – A Look Back In Time

Four phases of data collection using different

- Crawler implementations
- Approaches to find and sample (prioritize) seeds and URLs
- Page revisit policies

|             | crawler  | seeds / link prioritization | revisit policy                                   |
|-------------|----------|-----------------------------|--------------------------------------------------|
| 2008–2009   | Nutch    | list based                  |                                                  |
| 2012        | in-house | page rank                   |                                                  |
| 2013 – 2016 | Nutch    | seed donations, list based  |                                                  |
| 2017 – now  | Nutch    | harmonic centrality         | 30% new, 70% revisits<br>based on page relevance |

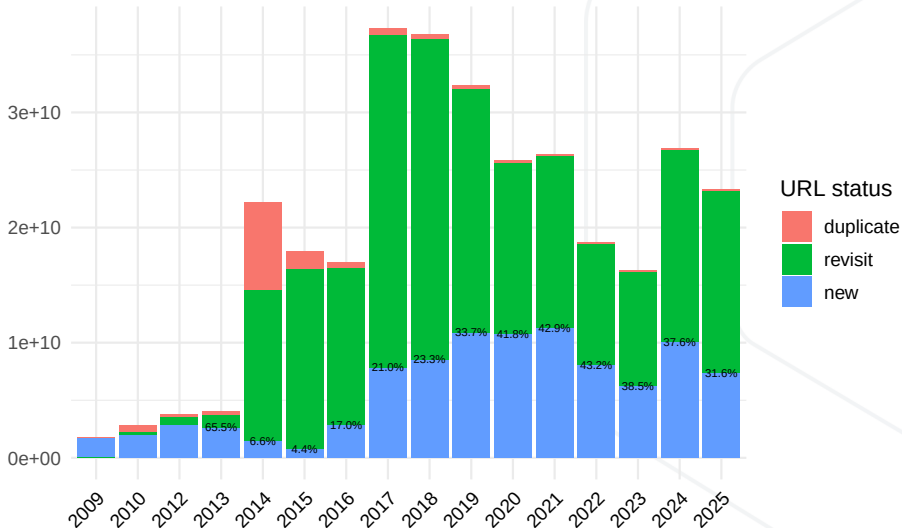
# What Is Representative?

Aspects of representativity:

- Breadth: coverage of unique domains (web sites)
- Depth: per-site coverage
- Freshness (new content)
- Amount of (near-)duplicates (per crawl and over multiple crawls)
- Regional coverage (top-level domains, content languages)
- Content quality
- Applicable for a given data use case?

# Size and Freshness

Number of Page Captures





# Crawler Politeness

- Crawl slowly
  - Further slow down (exponential backoff) if a site responds with errors
- Respect robots.txt rules (Robots Exclusion Protocol – [3]) and
- URI-level metatags (`<meta name=robots value=nofollow>`) [4, 5]
- CCBot identifies itself
  - User-agent string and contact information sent along with requests
  - Crawling from fixed list of IP addresses, publicly announced and verifiable via reverse DNS

Common Crawl – A Brief Introduction

## Data Pipelines

Prevalence of Web Data in LLMs

Anatomy of a Data Pipeline

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

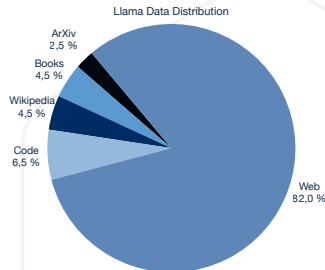
WMDQS

CommonLID Dataset

Summary

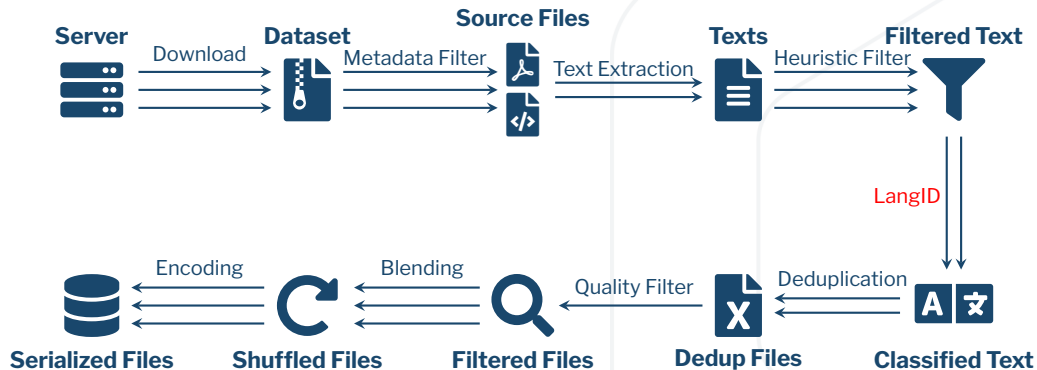
# Prevalence of Web Data in LLMs Today(ish)

- In practice datasets are not balanced
- Web data is always the cheapest and easiest to get
- Web data is diverse, but definitely not balanced or representative of all the language range.
- Web data always contains unwanted content (fiction, bias, propaganda).
- Programming language code (source code) is becoming ubiquitous in the data mix
- Public domain books and encyclopedic data is also common, but availability varies greatly between languages.



| LLaMA Data |            |
|------------|------------|
| Source     | Proportion |
| Web        | 82%        |
| Code       | 6.5%       |
| Wikipedia  | 4.5%       |
| Books      | 4.5%       |
| ArXiv      | 2.5%       |

# Anatomy of a Data Pipeline



**Figure 1:** A diagram of a common pre-processing pipeline. Arrows represent parallel processes; the more arrows there are, the more parallelizable the specific task is.

## A Survey of Pre-Processing Pipelines

- OSCAR (First use of LangID threshold for filtering) [6, 7]
- CCNet (Uses KenLM models [8] trained on Wikipedia) [9]
- mC4 (Uses word blocklists) [10]
- Refined Web (Uses Trafilatura for text extraction) [11]
- DataTrove (Uses Trafilatura and aggregates filters) [12]
- Dolma (Works on deduplication) [13]
- NeMo-Curator (Combines many heuristics and filters) [14]
- HPLT (Uses custom pipeline and has bilingual data) [15]
- MADLAD (Uses custom LangID and focuses on Multilinguality) [16]
- GlotCC (Uses Ungoliant [7] plus a new LangID model) [17]
- Community OSCAR (KenLM models trained on Adult Content) [18]
- And many others!

Common Crawl – A Brief Introduction

Data Pipelines

**Multilingual Data Pipelines**

Multilinguality is Hard

Plurality of Crawl is English

Language Identification Isn't Perfect

Current CCF LID: CLD2

Common Crawl's Multilingual Initiatives

WMDQS

CommonLID Dataset

Summary

# Multilinguality is Hard – Even with Web Data!

- The hope: lots of multilingual data on the web
- The reality: current pipelines struggle with multilinguality
  - **Plurality of crawl is English**
  - **Language identification isn't perfect**
  - Extraction and preprocessing not fully multilingual

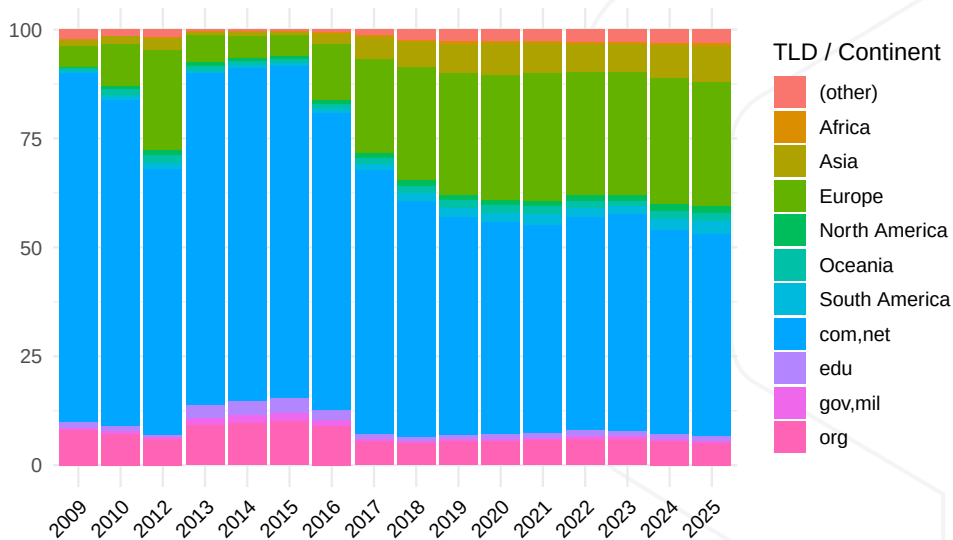
# Language Coverage Skews Towards English

|            | 2018         | 2019         | 2020         | 2021         | 2022         | 2023         | 2024         | 2025         |
|------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <other>    | 6.47         | 7.31         | 7.88         | 7.59         | 7.51         | 7.84         | 7.90         | 7.91         |
| <unknown>  | 3.32         | 2.62         | 2.35         | 2.53         | 2.80         | 2.84         | 3.00         | 2.91         |
| ara        | 0.76         | 0.59         | 0.56         | 0.60         | 0.63         | 0.64         | 0.66         | 0.67         |
| ces        | 1.03         | 1.04         | 1.05         | 1.06         | 1.03         | 1.09         | 1.04         | 1.02         |
| deu        | 5.15         | 5.47         | 5.57         | 5.63         | 5.60         | 5.79         | 5.36         | 5.57         |
| <b>eng</b> | <b>43.96</b> | <b>43.84</b> | <b>43.20</b> | <b>44.98</b> | <b>46.64</b> | <b>45.70</b> | <b>44.40</b> | <b>44.07</b> |
| fas        | 0.66         | 0.59         | 0.57         | 0.63         | 0.63         | 0.67         | 0.71         | 0.72         |
| fra        | 4.53         | 4.56         | 4.53         | 4.46         | 4.50         | 4.64         | 4.31         | 4.31         |
| ind        | 0.75         | 0.74         | 0.76         | 0.83         | 0.75         | 0.82         | 1.01         | 1.09         |
| ita        | 2.06         | 2.31         | 2.38         | 2.42         | 2.48         | 2.69         | 2.53         | 2.31         |
| jpn        | 5.47         | 4.81         | 4.78         | 4.66         | 4.68         | 4.83         | 5.03         | 5.09         |
| kor        | 0.59         | 0.69         | 0.76         | 0.66         | 0.65         | 0.67         | 0.71         | 0.77         |
| nld        | 1.50         | 1.74         | 1.79         | 1.80         | 1.93         | 2.09         | 1.85         | 1.81         |
| pol        | 1.68         | 1.70         | 1.68         | 1.62         | 1.62         | 1.70         | 1.80         | 1.75         |
| por        | 1.99         | 2.07         | 2.16         | 2.15         | 1.78         | 1.27         | 2.14         | 2.26         |
| <b>rus</b> | <b>9.27</b>  | <b>7.41</b>  | <b>7.11</b>  | <b>7.07</b>  | <b>5.90</b>  | <b>5.84</b>  | <b>6.02</b>  | <b>5.97</b>  |
| spa        | 4.18         | 4.16         | 4.25         | 4.33         | 4.36         | 4.58         | 4.56         | 4.46         |
| tur        | 0.90         | 0.87         | 0.91         | 1.00         | 0.86         | 0.84         | 1.18         | 1.15         |
| vie        | 0.75         | 0.74         | 0.80         | 0.92         | 0.92         | 1.04         | 1.01         | 1.03         |
| zho        | 4.98         | 6.76         | 6.91         | 5.07         | 4.73         | 4.41         | 4.75         | 5.12         |

Percentage of yearly crawls in selected languages, as identified by CLD2 Source: [19]



# Top-Level Domain Coverage Skews Western

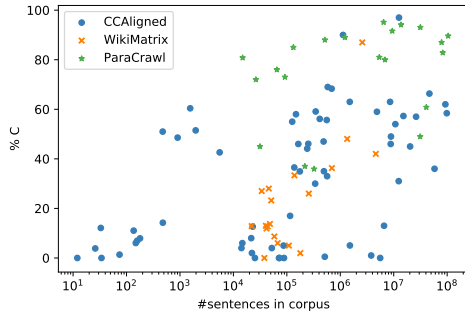
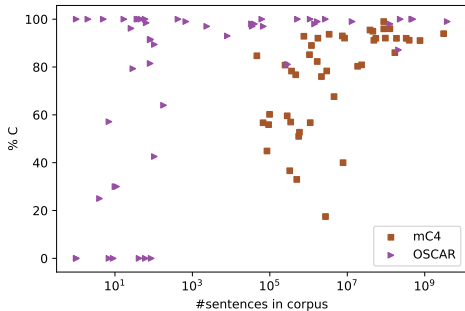


# Human Evaluation Shows Big Web LID Problems

|                        |           | Parallel    |                |            | Monolingual |             |
|------------------------|-----------|-------------|----------------|------------|-------------|-------------|
|                        |           | CCAligned   | ParaCrawl v7.1 | WikiMatrix | OSCAR       | mC4         |
| #langs audited / total |           | 65 / 119    | 21 / 38        | 20 / 78    | 51 / 166    | 48 / 108    |
| %langs audited         |           | 54.62%      | 55.26%         | 25.64%     | 30.72%      | 44.44%      |
| #sents audited / total |           | 8037 / 907M | 2214 / 521M    | 1997 / 95M | 3517 / 8.4B | 5314 / 8.5B |
| %sents audited         |           | 0.00089%    | 0.00043%       | 0.00211%   | 0.00004%    | 0.00006%    |
| macro                  | C         | 29.25%      | 76.14%         | 23.74%     | 87.21%      | 72.40%      |
|                        | X         | 29.46%      | 19.17%         | 68.18%     | -           | -           |
|                        | WL        | 9.44%       | 3.43%          | 6.08%      | 6.26%       | 15.98%      |
|                        | NL        | 31.42%      | 1.13%          | 1.60%      | 6.54%       | 11.40%      |
|                        | offensive | 0.01%       | 0.00%          | 0.00%      | 0.14%       | 0.06%       |
|                        | porn      | 5.30%       | 0.63%          | 0.00%      | 0.48%       | 0.36%       |
| micro                  | C         | 53.52%      | 83.00%         | 50.58%     | 98.72%      | 92.66%      |
|                        | X         | 32.25%      | 15.27%         | 47.10%     | -           | -           |
|                        | WL        | 3.60%       | 1.04%          | 1.35%      | 0.52%       | 2.33%       |
|                        | NL        | 10.53%      | 0.69%          | 0.94%      | 0.75%       | 5.01%       |
|                        | offensive | 0.00%       | 0.00%          | 0.00%      | 0.18%       | 0.03%       |
|                        | porn      | 2.86%       | 0.33%          | 0.00%      | 1.63%       | 0.08%       |
| #langs =0% C           |           | 7           | 0              | 1          | 7           | 0           |
| #langs <50% C          |           | <b>44</b>   | <b>4</b>       | <b>19</b>  | <b>11</b>   | <b>9</b>    |
| #langs >50% NL         |           | 13          | 0              | 0          | 7           | 1           |
| #langs >50% WL         |           | 1           | 0              | 0          | 3           | 4           |

Taken from [20]

# High-Resource $\neq$ High Quality



Taken from [20]

# CCF's Internal Eval (2018) => We Use CLD2

| Test Set                             | Languages  | Accuracy    |             |             |             | CPU Time<br>cumul. |
|--------------------------------------|------------|-------------|-------------|-------------|-------------|--------------------|
|                                      |            | dsl2014     | europarl    | twitter     | tatoeba     |                    |
| Docs                                 |            | 12600       | 21000       | 187461      | 19457       |                    |
| MiB                                  |            | 3.3         | 3.4         | 18.8        | 0.9         |                    |
| Languages                            |            | 10          | 21          | 137         | 143         |                    |
| cld2 [21]                            | 160        | .879        | .989        | .751        | .792        | 4                  |
| cld2 (Java bind) [21, 22, 23]        | 83         | .875        | .986        | .743        | .606        | 13                 |
| <b>cld2 (Java bind) [21, 22, 23]</b> | <b>160</b> | <b>.880</b> | <b>.990</b> | <b>.758</b> | <b>.792</b> | <b>14</b>          |
| cld2 Polyglot (Py bind) [24]         | 160        | .879        | .989        | .748        | .792        | 14                 |
| cld3 [25]                            | 101        | .830        | .991        | .595        | .630        | 29                 |
| fasttext [26]                        | 176        | .880        | .991        | .753        | .787        | 4                  |
| idNet [27]                           | 463        | .682        | .997        | .721        | .680        | 3:07:07            |
| langid (Python) [28]                 | 97         | .790        | .992        | .695        | .604        | 19:19              |
| langid (C) [29]                      | 97         | .790        | .992        | .695        | .604        | 7                  |
| langid (Java) [30]                   | 97         | .790        | .992        | .695        | .604        | 20                 |
| optimaize [31]                       | 71         | .740        | .969        | .534        | .443        | 1:25               |
| shuyo-cybozu [32]                    | 52         | .761        | .993        | .620        | .420        | 56                 |

## About CLD2

- Two variants: 83 or 160 supported languages
- Detects up to 3 languages per document
- Plain text or HTML input, valid UTF-8
- Quadgram Naïve Bayesian classifier with support of
  - Writing system / Unicode script
  - Ignoring web-specific character sequences (*http, copyright*)
  - Top-level domain of URL
  - Language codes in HTML and HTTP metadata
  - Character encoding (BIG5, KOI8-R)
- C++ library, trained on a proprietary corpus; no support for re-training

# Language Annotations in Common Crawl Data

- WARC metadata record

```
charset-detected: EUC-KR
languages-cld2: {
  "reliable":true,
  "text-bytes":2404,
  "languages":[
    {"code":"ko", "code-iso-639-3":"kor",
      "text-covered":0.97, "score":3735.0, "name":"Korean"},
    {"code":"en", "code-iso-639-3":"eng",
      "text-covered":0.02, "score":1264.0, "name":"ENGLISH"}]}
```

- WAT (same as in WARC metadata record)

- WET

WARC-Identified-Content-Language: kor,eng

- URL indexes: kor,eng (up to 3 language codes)

Common Crawl – A Brief Introduction

Data Pipelines

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

Improving Language Coverage and Balance

Initiative 1: The Web-Languages Project

Initiative 2: Improving LID for the Web

WMDQS

CommonLID Dataset

Summary

# Improving Language Coverage and Balance

Two main goals of current efforts:

- **Expand coverage** of crawl for languages other than English (LOTE)
- **Improve language identification** for web: more languages, better fidelity

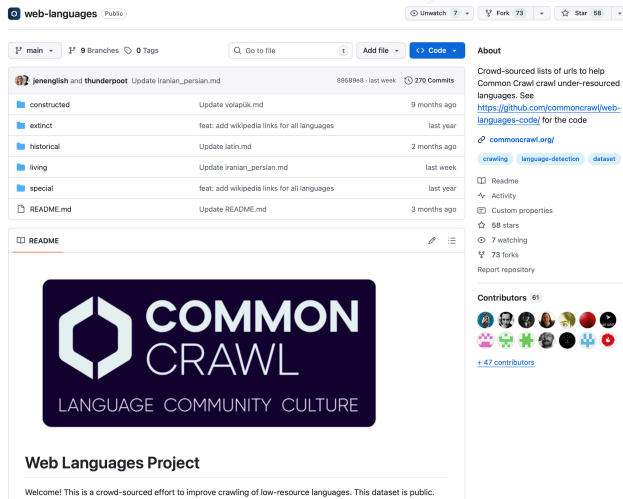


# Initiative 1: The Web-Languages Project

- Crowd-sourced effort to improve crawling of low-resource languages
- Language, Community, Culture
- Show the crawler high-quality links to “steer” into the right direction
- Hopefully, this will influence the harmonic centrality ranks over time

(<https://github.com/commoncrawl/web-languages>)

# The Web-Languages Project



The screenshot shows the GitHub repository page for `web-languages`. The repository is public and has 73 forks and 58 stars. The README file is selected, displaying the Common Crawl logo and the text "Web Languages Project". Below the logo, it says "Welcome! This is a crowd-sourced effort to improve crawling of low-resource languages. This dataset is public." The file list on the left shows various language-specific directories and files, including `constructed`, `extinct`, `historical`, `living`, `special`, and `README.md`.

**web-languages** Public

Unwatch 7 Fork 73 Star 58

main 9 Branches 0 Tags Go to file Add file Code

**jenenglish and thunderpoot** Update `iranian_persian.md` 89589e8 · last week 270 Commits

| File                     | Description                                 | Last Commit  |
|--------------------------|---------------------------------------------|--------------|
| <code>constructed</code> | Update <code>volapük.md</code>              | 9 months ago |
| <code>extinct</code>     | feat: add wikipedia links for all languages | last year    |
| <code>historical</code>  | Update <code>latin.md</code>                | 2 months ago |
| <code>living</code>      | Update <code>iranian_persian.md</code>      | last week    |
| <code>special</code>     | feat: add wikipedia links for all languages | last year    |
| <code>README.md</code>   | Update <code>README.md</code>               | 3 months ago |

**README**

  
**COMMON CRAWL**  
LANGUAGE COMMUNITY CULTURE

**Web Languages Project**

Welcome! This is a crowd-sourced effort to improve crawling of low-resource languages. This dataset is public.

**About**

Crowd-sourced lists of urls to help Common Crawl crawl under-resourced languages. See <https://github.com/commoncrawl/web-languages-code/> for the code

[commoncrawl.org/](https://commoncrawl.org/)

[crawling](#) [language-detection](#) [dataset](#)

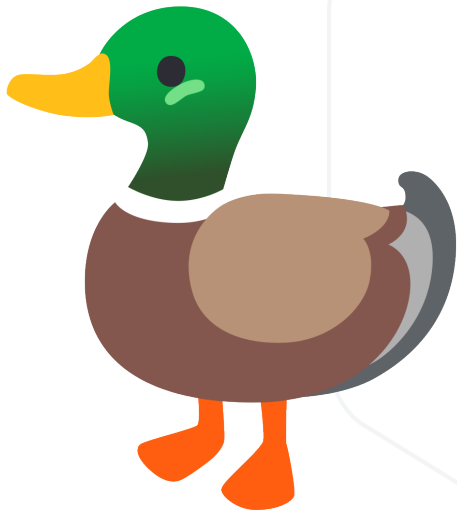
Readme Activity Custom properties 58 stars 7 watching 73 forks Report repository

**Contributors** 61

+ 47 contributors

**Figure 2:** <https://github.com/commoncrawl/web-languages>

## Initiative 2: Improving Language Identification for the Web



Common Crawl – A Brief Introduction

Data Pipelines

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

## WMDQS

- The LangID Project

- Why LangID

- The Annotation Platform

- The Incentives to Annotate

- The Organizations Behind the Effort

- The Hackathons

- WMDQS

- The Results

CommonLID Dataset

Summary

# The LangID Project

The screenshot shows the LangID project interface on the MLCommons Dyna Bench platform. The top navigation bar includes the MLCommons logo, 'About', 'Communities', a language selector set to 'English', and a search bar. The main header area features a colorful background with text and symbols, displaying 'Text Language Identification', 'Common Crawl's Lang ID', '1 ROUNDS', and '29636 EXAMPLES'. Below this, there are tabs for 'Leaderboard' and 'Overview', and a 'Create Examples' button. The 'Description' tab is active, showing the task title 'Common Crawl - MLCommons Language Identification task'. The 'Instructions' section welcomes users and explains the goal of creating a new LangID dataset from Common Crawl data. It also provides a prompt for annotators to select a language they are proficient in. At the bottom, there is a text input field with the placeholder 'Please select a language you are proficient in'.

MLCommons Dyna Bench

About Communities

English Search

Others

Text Language Identification

Common Crawl's Lang ID

1 ROUNDS

29636 EXAMPLES

Leaderboard Overview

Create Examples

Description

## Common Crawl - MLCommons Language Identification task

### Instructions

Welcome to the Language Identification task from Common Crawl and MLCommons!

The main goal of our task is to produce a new LangID dataset solely based on Common Crawl's data that covers as many languages as possible, with the aim of improving our LangID model so that we can discover more content for your language.

In this task, annotators will be first give a prompt in which they select a language that they are proficient on. The bar is a search field so that the annotator can easily find the language they are looking for:

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

Please select a language you are proficient in

**Figure 3:** Now closed

## Why LangID

- Existing models are old and unmaintained
- LangID models keep being used as filters but nobody studies the impact of this
- High coverage models remain largely under-evaluated due to lack of data
- The question of generalization across domains is under-studied
  - Most LangID datasets contain very clean data
- We want to expand the language coverage of Common Crawl and we need better signals
- LangID models need to accurately classify noisy web text
- Crowd-sourced data annotation task to collect manual annotations of web data

# The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

9 examples created ? 0

Please select a language you are proficient in

Select a language

Select

**Figure 4:** A screenshot of the annotation platform interface. The participant selects the language they want to annotate for.

# The Annotation Platform

TEXT LANGUAGE IDENTIFICATION  
Label the text with the languages you think it is written in

43 examples created

To tag text in an additional language, select that language from the dropdown, then select the text. To undo a language tag, click anywhere in the selected text.

Select all text area

Spanish; Castilian

1: spa 2: eng 3: fra

ESTUDIO DE LA EXPRESIÓN Y FUNCIÓN DE RECEPTORES DE APOLIPOPROTEÍNA E DURANTE LA DIFERENCIACIÓN DE UN MODELO NEURONAL

Resumen | Convocatoria | Responsable

Resumen

Las Entidades Neurodegenerativas tales como la enfermedad de Parkinson (EP) y enfermedad de Alzheimer (EA) constituyen las dos primeras entidades degenerativas del SNC, y por tanto son parte frecuente de la consulta especializada de Neurología. En los últimos años se han realizado muchos esfuerzos para determinar los factores tanto genéticos como ambientales que pueden llevar al estado patológico. Asimismo, los trastornos del desarrollo del sistema nervioso central son preocupación constante en la clínica. Ambos grupos de entidades, al parecer confluyen en una vía común que está muy relacionada con la señalización dependiente de los lípidos. El objetivo de este proyecto es esclarecer el papel de la proteína Apolipoproteína E (APOE) y sus receptores en el balance muerte/supervivencia celular estudiando sus efectos en un modelo neuronal. La primera fase del proyecto consiste en observar los niveles de expresión de cuatro de las proteínas que actúan como receptores para APOE, entre ellas LDLR (Low Density Lipoprotein Receptor), LRP1 (Lipoprotein Receptor Related Protein-1), VLDLR (Very Low Density Lipoprotein Receptor) y APOER2 (Apolipoprotein E Receptor 2) durante el periodo de diferenciación de la línea celular CAD (Cath.a differentiated). En la segunda fase, se agregará proteína APOE recombinante al medio de cultivo para analizar sus consecuencias en el número y viabilidad de las células en un contexto neurotóxico inducido por ceramida; y posteriormente se analizará sus efectos en vías de señalización cinasa de tirosina, específicamente la vía PI3K/AKT junto con sus componentes corriente abajo, así como la activación de vías de muerte celular como las caspasas.

Convocatoria

Nombre de la convocatoria: FACULTAD DE MEDICINA 2008 - CONVOCATORIA PARA EL ESTÍMULO A LA INVESTIGACIÓN A TRAVÉS DE PROYECTOS Y ENFOQUES ESTRATÉGICOS, DE PRIORIDADES E INTERDISCIPLINARIOS

Modalidad: Modalidad I: Apoyo a Grupos de Investigación de la Facultad de Medicina, a través de proyectos de investigación - Grupos Categorizados por COLCIENCIAS como C y los demás grupos de investigación de la Facultad de Medicina. [ver](#)

Optional annotations: Choose one or more of the following tags for the document (only if they apply).

☐ Sexual The text contains explicit sexual or adult content.

☐ Toxic The snippet contains offensive or harmful content.

☐ Non-Standard Orthography The snippet contains variations in spelling, punctuation, and written conventions that deviate from established language norms.

☐ Boilerplate The text snippet contains linguistic content, but most of the content (more than 50%) is part of a template, form, standard alert text, a menu, etc.

Submit

Skip and load a new text

Figure 5: The participant has highlighted all text as Spanish.



# The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

43 examples created ? ⓘ

To tag text in an additional language, select that language from the dropdown, then select the text. To undo a language tag, click anywhere in the selected text.

Select all text area

English ▼ 1: spa 2: eng 3: fra

Esta sección trata los aspectos relacionados con la jugabilidad de las partidas, como el uso de determinado software, la manera de pilotar tu nave en diferentes situaciones, y consejos para gestionar tu imperio. Los juegos están organizados en secciones, así que asegúrate de que el juego para el que estás mirando la ayuda está seleccionado en la barra de la izquierda.

Esta sección contiene preguntas que no pertenecen a una categoría determinada o que son de interés general. »»

[EN] My gameplay question is not listed here. – what do I do? eng

[ES] Puedo aterrizar en planetas spa

[EN] My reputation with the Paranid is Enemy of Priest Duke 18%, and when I kill enemies it changes to 19%. Is this correct? eng

— Optional annotations: Choose one or more of the following tags for the document (only if they apply).

☐ Sexual The text contains explicit sexual or adult content.

☐ Toxic The snippet contains offensive or harmful content.

☐ Non-Standard Orthography The snippet contains variations in spelling, punctuation, and written conventions that deviate from established language norms.

☐ Boilerplate The text snippet contains linguistic content, but most of the content (more than 50%) is part of a template, form, standard alert text, a menu, etc.

Submit Skip and load a new text

**Figure 6:** A screenshot of the annotation platform interface. The participant has highlighted the English and Spanish text in the extract in different colours.

# The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

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[ES] Puedo aterrizar en planetas spa

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☐ Boilerplate The text snippet contains linguistic content, but most of the content (more than 50%) is part of a template, form, standard alert text, a menu, etc.

Submit Skip and load a new text

**Figure 7:** A screenshot of the annotation platform interface. The participant has highlighted the English and Spanish text in the extract in different colours.

# The Incentives to Annotate

- Better LangID performance for your language!
- Contributors who annotated at least 100 documents are invited to be authors on the dataset paper
- Encourage the development of open-source LangID models
- More and higher-quality (and noisy!) multilingual data, especially for low-resource languages

# The Organizations Behind the Effort



# Masakhane Hackathon

## AfriCC

African Common Crawl Datasets



### URL Collection

Browse about the curated collection of African cultural URLs



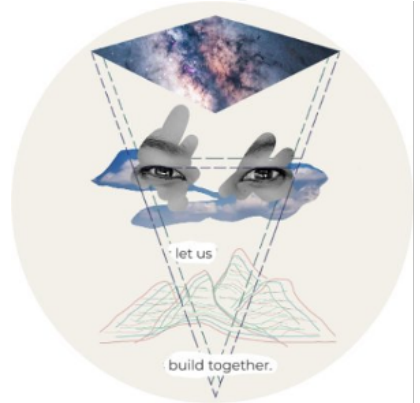
### LID Annotation

Explore language identification annotations datasets across African languages



### Contribute?

Interested in contributing to the AfriCC project? Join us!



## Masakhane

# Masakhane Hackathon

## AfriCC LID Instances per Language

Comprehensive visualization of African language representation in the dataset

63

African Languages

9,698

Total Entries

154

Average per Language

### All African Languages

Search languages...

Sort A-Z

Sort by Region

Sort by Instances

| Language ↑           | Region         | Instances |
|----------------------|----------------|-----------|
| Acoli                | East Africa    | 1         |
| Aja (Benin)          | West Africa    | 3         |
| Amharic              | Horn of Africa | 574       |
| Antankarana Malagasy | Malagasy       | 1         |
| Bambara              | West Africa    | 15        |

zoom.us

# SEACrowd Hackathon



The poster features a light gray background with a subtle circuit pattern. At the top, the SEACrowd logo is displayed in a large, bold, black font, with the 'SEA' part in a colorful, stylized font. Below the logo, the text 'X' is centered. Underneath, three logos are arranged horizontally: COMMON CRAWL, ML Commons, and EleutherAI. The main title 'Language Identification Hackathon' is written in a bold, red font. Below this, there are two purple rounded rectangles. The left one contains the 'When:' section with a clock icon and a list of time slots for different time zones. The right one contains the 'Where:' section with a location pin icon, the text 'Google Meet', and a QR code. At the bottom, there are two more QR codes: one for 'Hear about future events Join the Discord!' with a Discord icon, and another for 'Annotation Platform' with a small icon.

**SEACrowd**

X

COMMON CRAWL ML Commons ELEUTHERAI

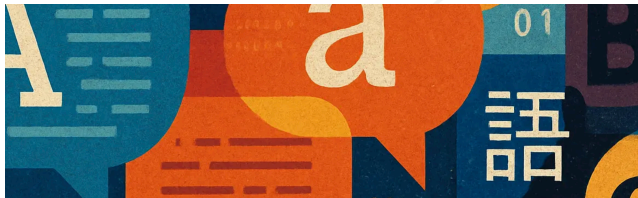
**Language Identification Hackathon**

**When:**  
July 22, 2025  
8-11am UTC -4 (US East Coast)  
1-4pm UTC +1 (UK)  
2-5pm UTC +2 (Central Europe)  
7-10pm UTC +7 (Indochina Time)

**Where:**  
Google Meet

Hear about future events  
Join the Discord!

Annotation Platform



## 1st Workshop on Multilingual Data Quality Signals

Palais des Congrès  
Montréal, Canada  
10 October 2025

[Call for Papers](#) [Shared Task](#)

Recent research has shown that large language models (LLMs) not only need large quantities of data, but also need data of sufficient quality. Ensuring data quality is even more important in a multilingual setting, where the amount of acceptable training data in many languages is limited. Indeed, for many languages even the fundamental step of language identification remains a challenge, leading to unreliable language labels and thus noisy datasets for downstream language

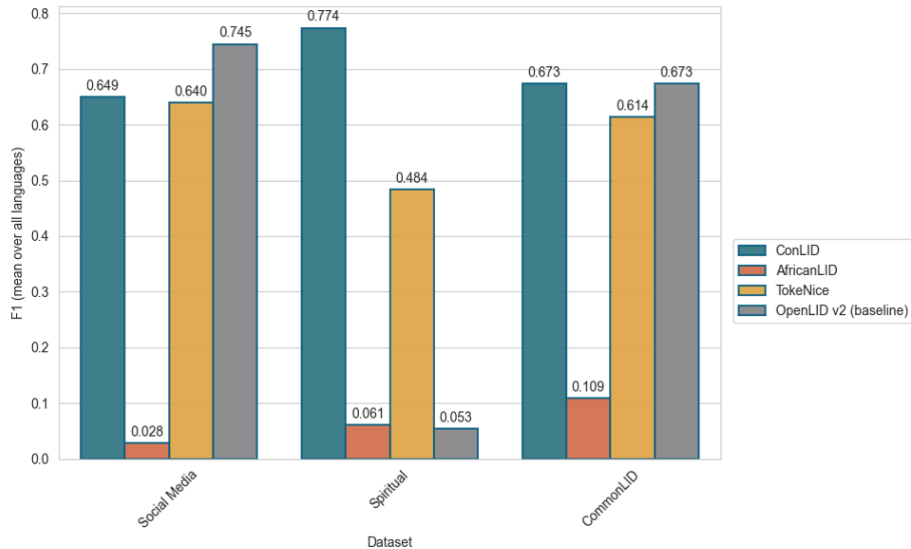
**Figure 8:** <https://wmdqs.org>



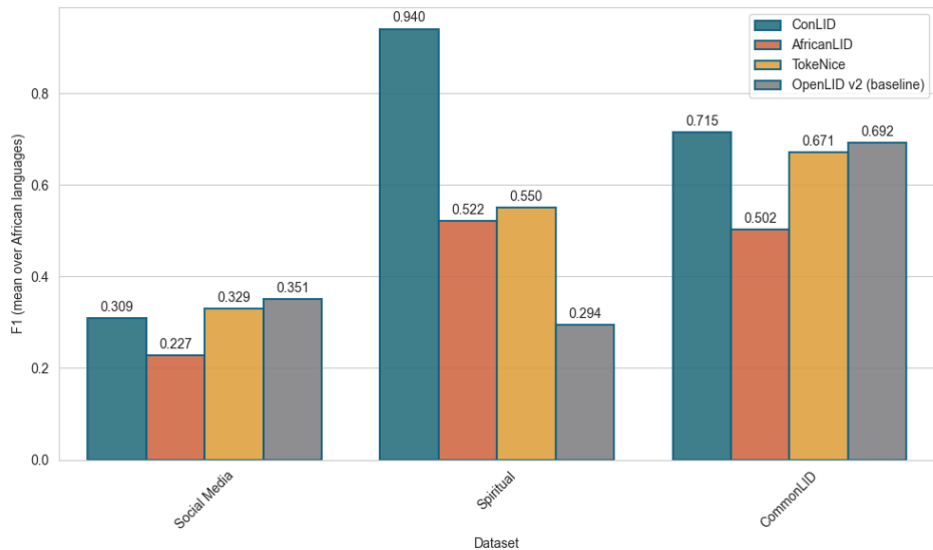
## CommonLID v0.1 and a LangID Shared Taks

- CommonLID: 22K documents annotated or 200k annotated lines
  - 62 languages with >100 annotated lines
  - 42 languages with >500 annotated lines
  - 24 languages with >1000 annotated lines
  - 8 languages with >10000 annotated lines
- WMDQS LangID Shared Task
  - Evaluated on Dynabench
  - 18 different teams got into the shared task
  - 4 teams submitted models
  - 3 models ran on Dynabench
  - 3 Benchmarks were used: Social media dataset, spiritual text dataset and CommonLID

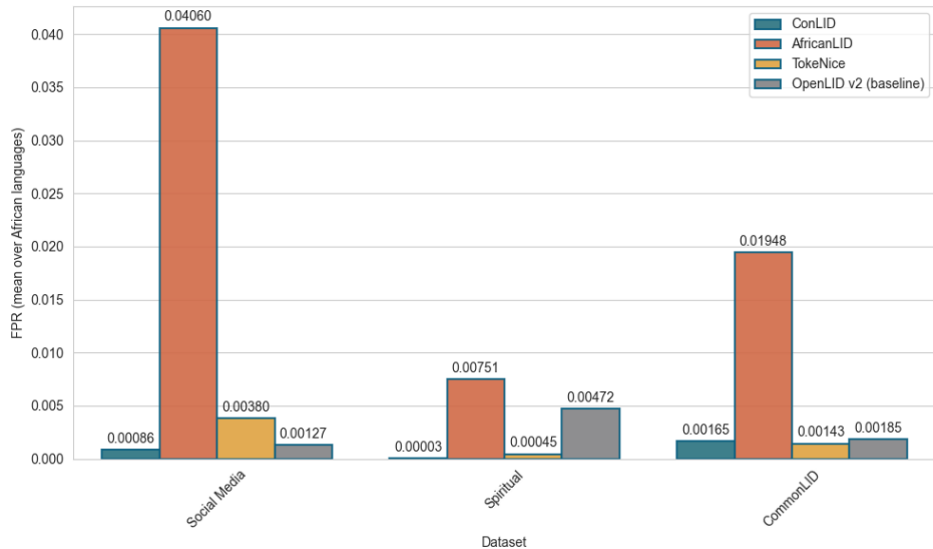
# F1 Results: All Languages



# F1 Results: African Languages



# False-positive-rate: African Languages



Common Crawl – A Brief Introduction

Data Pipelines

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

WMDQS

CommonLID Dataset

- Overview

- Creation

- Evaluation

Summary

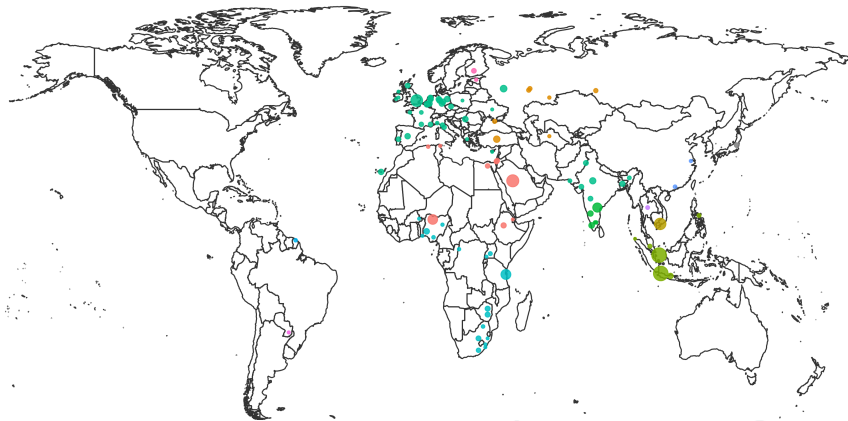
# Overview of CommonLID

Annotated data => LID evaluation dataset

- 373,230 lines
- 109 languages
- 78 languages with >100 lines
- 1-43,189 lines per class (mean 3424)
- 215.5 mean characters per line

All participants who annotated > 100 lines were offered co-authorship.

# Geographic distribution of CommonLID



**Figure 9:** Illustrative geographical distribution of the language varieties in CommonLID. The dot size represents the number of annotated lines, and colour represents language family.

# Creating CommonLID: Preprocessing

Converting to line-level annotations not trivial!

- Split on new lines
- Strip leading/trailing whitespace, de-duplicate line+language pairs
- Fuzzy match lines with multiple annotations



# Creating CommonLID: Multiple Labels

Reconciling labels was a combination of automatic and manual processes

- Filter out short lines (< 10 characters)
- English contamination: `langid.py`, filter (most) lines  $P(eng) > 0.7$
- Manual inspection of remaining multiple labels
  - Most remaining were clear mistakes => remove
  - Kept macro/micro language labels (both potentially valid)
  - Kept lines with multiple languages (none dominant)
- All languages audited (three spurious language classes removed)

# Evaluating CommonLID

We evaluate 8 popular LID models using CommonLID + 5 evaluation datasets.

Fair comparison isn't easy:

- Label normalisation between eval sets and between model outputs (iso639-lang)
- Different models cover different languages: bigger not always better
- Web scale LID needs speed *and* fidelity!

# Summary of Evaluated LID Models

| Name             | # langs. | Architecture         | Data sources                    | Open data? |
|------------------|----------|----------------------|---------------------------------|------------|
| AfroLID [33]     | 517      | Transformer          | ≈ 100M curated sents.           | ✗          |
| CLD2 [21]        | 158      | Naïve Bayes          | Web pages (curated and scraped) | ✗          |
| fasttext [34]    | 218      | FastText             | “publicly available datasets”   | ✗          |
| FUN-LangID [35]  | 1634     | Common sub-strings   | Web+Wikipedia+Bibles            | ✗          |
| pyFranc [36]     | 414      | Trigram distribution | UDHR                            | ✓          |
| gCLD3 [37]       | 99       | Neural network       | ?                               | ✗          |
| GlottLID v3 [38] | 1868     | FastText             | Curated open sources            | ✓          |
| OpenLID-v2 [39]  | 193      | FastText             | Curated, audited open sources   | ✓          |

**Table 1:** Summary of LID models used.

## Overlap in LID Model Coverage

|            | GlottLID    | FUN-LangID  | AfroLID    | pyFranc    | fasttext   | OpenLID-v2 | CLD2       | gCLD3     |
|------------|-------------|-------------|------------|------------|------------|------------|------------|-----------|
| GlottLID   | <b>1868</b> |             |            |            |            |            |            |           |
| FUN-LangID | 1351        | <b>1549</b> |            |            |            |            |            |           |
| AfroLID    | 401         | 313         | <b>515</b> |            |            |            |            |           |
| pyFranc    | 362         | 312         | 93         | <b>410</b> |            |            |            |           |
| fasttext   | 202         | 180         | 49         | 166        | <b>210</b> |            |            |           |
| OpenLID-v2 | 190         | 157         | 47         | 156        | 180        | <b>193</b> |            |           |
| CLD2       | 128         | 153         | 30         | 121        | 114        | 104        | <b>158</b> |           |
| gCLD3      | 83          | 99          | 13         | 80         | 79         | 78         | 98         | <b>99</b> |

**Table 2:** Counts of mutual language coverage between LID models (models in descending order by coverage).

# Macro-average F1 by model/evaluation set

|    | FLORES+<br>209 languages |                              | SmolSent<br>82 languages |                             | UDHR-LID<br>418 languages |                              | Bibles<br>1047 languages |                              | Social Media<br>66 languages |                            | CommonLID<br>109 languages |                             |
|----|--------------------------|------------------------------|--------------------------|-----------------------------|---------------------------|------------------------------|--------------------------|------------------------------|------------------------------|----------------------------|----------------------------|-----------------------------|
|    | all                      | cov.                         | all                      | cov.                        | all                       | cov.                         | all                      | cov.                         | all                          | cov.                       | all                        | cov.                        |
| AL | 15.9                     | 67.9 <sub>(49)</sub>         | 42.3                     | 69.4 <sub>(50)</sub>        | 15.2                      | 67.6 <sub>(94)</sub>         | 12.1                     | 69.8 <sub>(181)</sub>        | 1.5                          | <b>98.0</b> <sub>(1)</sub> | 9.2                        | 43.5 <sub>(23)</sub>        |
| C2 | 45.0                     | 87.8 <sub>(105)</sub>        | 40.6                     | 83.2 <sub>(40)</sub>        | 24.0                      | 81.7 <sub>(121)</sub>        | 4.4                      | 73.8 <sub>(62)</sub>         | 76.8                         | 85.9 <sub>(59)</sub>       | 49.5                       | <b>79.3</b> <sub>(68)</sub> |
| FT | 80.3                     | 92.2 <sub>(182)</sub>        | 48.1                     | 83.8 <sub>(47)</sub>        | 27.0                      | 68.2 <sub>(166)</sub>        | 3.3                      | 43.8 <sub>(79)</sub>         | <b>83.6</b>                  | 84.9 <sub>(65)</sub>       | 49.3                       | 72.6 <sub>(74)</sub>        |
| FL | 68.8                     | 86.2 <sub>(165)</sub>        | <b>75.9</b>              | 79.8 <sub>(78)</sub>        | 59.0                      | 78.2 <sub>(314)</sub>        | 71.3                     | 89.1 <sub>(837)</sub>        | 58.7                         | 65.5 <sub>(58)</sub>       | 46.4                       | 57.8 <sub>(86)</sub>        |
| Fr | 61.9                     | 80.4 <sub>(160)</sub>        | 46.4                     | 76.0 <sub>(50)</sub>        | <b>93.4</b>               | <b>95.9</b> <sub>(407)</sub> | 7.6                      | 52.1 <sub>(152)</sub>        | 62.4                         | 66.4 <sub>(62)</sub>       | 39.3                       | 61.3 <sub>(70)</sub>        |
| C3 | 28.1                     | 71.5 <sub>(78)</sub>         | 11.9                     | 57.4 <sub>(17)</sub>        | 11.0                      | 54.4 <sub>(80)</sub>         | 2.1                      | 42.8 <sub>(47)</sub>         | 63.1                         | 75.7 <sub>(54)</sub>       | 34.3                       | 66.2 <sub>(55)</sub>        |
| GL | <b>94.2</b>              | <b>96.5</b> <sub>(204)</sub> | 74.7                     | <b>91.4</b> <sub>(67)</sub> | 73.7                      | 84.4 <sub>(366)</sub>        | <b>82.3</b>              | <b>93.0</b> <sub>(927)</sub> | 80.6                         | 85.8 <sub>(62)</sub>       | <b>60.4</b>                | 68.6 <sub>(96)</sub>        |
| OL | 83.5                     | 90.4 <sub>(193)</sub>        | 45.4                     | 82.7 <sub>(45)</sub>        | 25.1                      | 67.3 <sub>(156)</sub>        | 3.3                      | 44.6 <sub>(78)</sub>         | 83.4                         | 88.7 <sub>(62)</sub>       | 47.4                       | 68.0 <sub>(76)</sub>        |

**Table 3:** AfroLID (AL), CLD2 (C2), fasttext (FT), FUN-LangID (FL), pyFranc (Fr), CLD3 (C3), GlotLID (GL), OpenLID-v2 (OL). Scores over whole dataset (*all*) and on subset of covered languages (*cov.*). (x) = Count of languages in the eval. set covered by model.

## Performance on Common Subsets of Languages

|    | High-coverage<br>1351 languages |               | Afro*<br>294 languages |              | Core<br>76 languages |             |
|----|---------------------------------|---------------|------------------------|--------------|----------------------|-------------|
|    | F1 ↑                            | FPR ↓         | F1 ↑                   | FPR ↓        | F1 ↑                 | FPR ↓       |
| GL | <b>87.8</b>                     | <b>0.006%</b> | <b>88.7</b>            | <b>0.02%</b> | <b>91.6</b>          | <b>0.1%</b> |
| FL | 76.3                            | 0.02%         | 73.5                   | 0.08%        | 77.9                 | 0.2%        |
| AL | —                               | —             | 71.8                   | 0.7%         | —                    | —           |
| Fr | —                               | —             | —                      | —            | 77.1                 | 0.4%        |
| FT | —                               | —             | —                      | —            | 77.7                 | 0.5%        |
| OL | —                               | —             | —                      | —            | 78.5                 | 0.5%        |
| C2 | —                               | —             | —                      | —            | 84.7                 | 0.2%        |
| C3 | —                               | —             | —                      | —            | 61.6                 | 1.4%        |

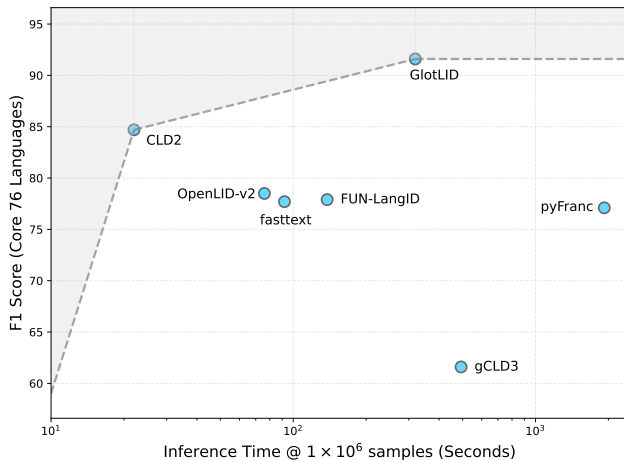
**Table 4:** Performance on combined test data. GlotLID (GL), FUN-LangID (FL), AfroLID (AL), pyFranc (Fr), fasttext (FT), OpenLID-v2 (OL), CLD2 (C2), CLD3 (C3).

# Inference Speed Varies Greatly by Model

| Model      | Samples/s | Duration @1e + 6 samples |
|------------|-----------|--------------------------|
| CLD2       | 43735     | 22s                      |
| OpenLID-v2 | 13123     | 1m 16s                   |
| fasttext   | 10867     | 1m 32s                   |
| FUN-LangID | 7237      | 2m 18s                   |
| GlottLID   | 3127      | 5m 19s                   |
| gCLD3      | ‡2026     | 8m 13s                   |
| Franc      | 520       | 32m 03s                  |
| AfroLID    | †66       | 4h 12m 49s               |

**Table 5:** Inference speed for LID models on FLORES+. Inference done on a 14-core Apple M4 Pro chip with 64GB of RAM. †AfroLID uses transformers - performance is likely better on CUDA hardware. ‡Measured on an AMD EPYC 7351P (16 cores @ 2.4GHz) Linux machine with 256GB RAM as gCLD3 not usable on modern macOS.

# Trade Off Between Inference Speed and F1



**Figure 10:** Inference speed vs. F1 (76 core languages, combined datasets)



## GlottLID Beats LLM Performance for LID

|                          | High-cov.<br>1351 langs. |       | Afro*<br>294 langs. |       | Core<br>76 langs. |       |
|--------------------------|--------------------------|-------|---------------------|-------|-------------------|-------|
|                          | F1 ↑                     | FPR ↓ | F1 ↑                | FPR ↓ | F1 ↑              | FPR ↓ |
| GlottLID                 | 90.0                     | 0.01% | 90.6                | 0.05% | 93.5              | 0.13% |
| <i>OpenAI GPT models</i> |                          |       |                     |       |                   |       |
| 4o-mini                  | 48.5                     | 0.09% | 43.6                | 0.69% | 81.0              | 0.54% |
| 4o                       | 62.0                     | 0.07% | 57.3                | 0.46% | 89.0              | 0.33% |
| 5-mini                   | 60.4                     | 0.06% | 55.1                | 0.43% | 76.4              | 0.53% |
| 5                        | 70.3                     | 0.05% | 66.6                | 0.29% | 91.8              | 0.22% |

**Table 6:** A comparison of large language models from OpenAI’s GPT family (4o-mini, 4o, 5-mini, 5) and GlottLID (best performing LID model). We report performance on the combined but down-sampled test data (15k test samples), subsets as in .

# ArXiv Paper



<https://www.arxiv.org/abs/2601.18026>

Common Crawl – A Brief Introduction

Data Pipelines

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

WMDQS

CommonLID Dataset

**Summary**

# Summary

- Multilinguality is hard!
- Language Identification is hard!
  - Comparing and evaluating models is very difficult
  - Assessing language coverage is complicated, especially when different language registers are considered
  - We need more community initiatives to build better datasets
  - The Common Crawl Foundation will continue working on these language initiatives, but we need your help!
  - We need to develop better architectures and ensure long-term maintenance

# Questions?



<https://github.com/commoncrawl/presentations>

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